

Col-2 Assessment-2

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Scenario-Based Questions

1. Crack Detection on Concrete Surfaces Using Drone Images

A. Image Preprocessing Pipeline

To detect cracks accurately, the drone images should be cleaned and enhanced before edge detection. A suitable preprocessing pipeline is:

Step 1: Image Acquisition

- Capture high-resolution images using a drone.
- Use image stabilization if possible to reduce blur caused by camera movement.

Why?

A clear image provides better information for crack detection.

Step 2: Illumination Correction

- Apply **Homomorphic Filtering** or **CLAHE (Contrast Limited Adaptive Histogram Equalization)**.

Why?

Outdoor images often contain uneven lighting due to sunlight and shadows. Illumination correction balances the brightness across the image so that cracks become more visible.

Step 3: Noise Reduction

- Use a **Gaussian Filter** or **Bilateral Filter**.

Why?

Drone images usually contain Gaussian noise caused by the camera sensor or vibration. Noise reduction removes unwanted pixels while preserving the crack edges.

Step 4: Contrast Enhancement

- Apply **CLAHE**.

Why?

Some cracks are very thin and difficult to see. Contrast enhancement makes these cracks stand out clearly from the concrete surface.

Step 5: Image Normalization

- Normalize pixel intensity values between **0–255** or **0–1**.

Why?

Normalization gives all images a consistent intensity range, making the edge detection process more reliable.

Final Pipeline

Image Acquisition → Illumination Correction → Noise Reduction → Contrast Enhancement → Image Normalization → Edge Detection

This sequence improves image quality and increases the accuracy of crack detection.

B. Suitable Edge Detection Technique

The **Canny Edge Detector** is the best choice for detecting thin cracks.

Reasons

- Detects very fine and weak edges.
- Removes noise before detecting edges.
- Produces thin and continuous edge lines.
- Reduces false edge detection.

Comparison with Sobel

Canny Edge Detector	Sobel Edge Detector
Excellent noise removal	Limited noise removal
Detects thin cracks accurately	May miss fine cracks
Produces continuous edges	Produces thicker edges
Fewer false edges	More false edges

Conclusion: Since cracks are thin and irregular, the Canny edge detector gives more accurate and reliable results than the Sobel operator.

C. Post-Processing Technique

After edge detection, some crack segments may appear broken.

A suitable solution is **Morphological Closing** (Dilation followed by Erosion).

Benefits

- Connects broken crack segments.
- Fills small gaps.
- Produces continuous crack lines.

If required, **Skeletonization** can also be applied to obtain single-pixel-wide crack lines for accurate measurement.

2. Human Motion Tracking in a Security Camera Video

A. Lucas–Kanade Optical Flow

Lucas–Kanade Optical Flow estimates the movement of objects between two consecutive video frames.

Working Process

1. Detect feature points such as corners.
2. Assume the brightness of each point remains constant.
3. Calculate the movement of each point between consecutive frames.
4. Generate motion vectors showing the direction and speed of movement.

These motion vectors help track people as they walk through the hallway.

B. Aperture Effect

The **Aperture Effect** occurs when an object is viewed through a small region, making it difficult to determine its actual direction of motion.

In areas such as:

- Plain walls
- Smooth floors
- Simple clothing

there is very little texture. As a result, the optical flow algorithm cannot estimate motion accurately.

This may lead to incorrect motion vectors or even failure to detect movement.

C. Sudden Change in Speed or Direction

If a person suddenly changes speed or direction:

- The optical flow vectors also change in length and direction.
- Tracking may become unstable for a few frames.

Improvement Method

Use **Pyramidal Lucas–Kanade Optical Flow**.

It estimates motion at multiple image scales, making it more effective for handling large or sudden movements.

Additional techniques such as the **Kalman Filter** can also predict future positions and improve tracking accuracy.

3. Vehicle Tracking at a Traffic Junction

A. Optical Flow for Vehicle Tracking

Optical flow measures how image pixels move between consecutive video frames.

Working Steps

1. Detect moving vehicles.
2. Calculate motion vectors.
3. Estimate the direction of movement from the vector orientation.
4. Estimate vehicle speed from the amount of pixel displacement over time.

This information is useful for vehicle tracking, traffic monitoring, and speed estimation.

B. Challenges in Motion Estimation

Challenge 1: Occlusion

When two vehicles overlap, one vehicle may partially or completely hide another.

Effect

- Some feature points disappear.
- Optical flow cannot estimate motion correctly.
- Vehicle tracking may temporarily fail.

Challenge 2: Shadows

Moving shadows can be mistaken for moving vehicles.

Effect

- Creates incorrect motion vectors.
- Produces false detections.
- Reduces tracking accuracy.

Other challenges include sudden braking, sharp turns, and camera vibration, all of which can affect motion estimation.

C. Improvement Method

A reliable solution is to combine **YOLO object detection**, **DeepSORT tracking**, and a **Kalman Filter**.

Why this combination works

- **YOLO** accurately detects vehicles in each frame.
- **DeepSORT** keeps track of each vehicle even when they overlap.
- **Kalman Filter** predicts the next position of a vehicle, allowing tracking to continue during short occlusions or sudden changes in direction.

This combination provides accurate and robust vehicle tracking in busy traffic conditions.